

Section 12.5: Equations of Lines and Planes

Vectors and the Geometry of Space

MTH 310
Calculus III

Lines in 2D — Review

Example 1

Find the equation of the line through $P(0, 0)$ and $Q(1, 2)$.

Find the slope, then use point-slope form.

Parametric Equations of Lines

Example 2

Write parametric equations for the line through $P(0, 0)$ with direction $\langle 1, 2 \rangle$.

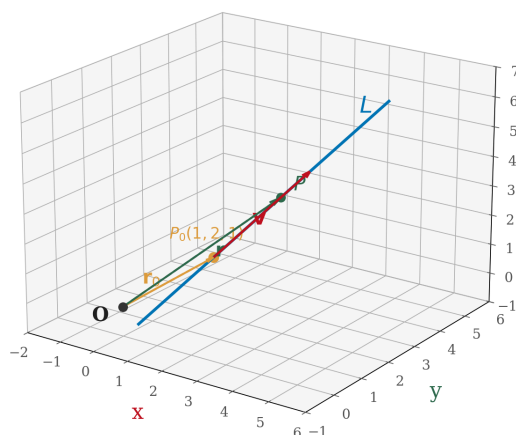
Use the parametric form with the given point and direction.

Vector Equation of a Line

Vector Equation of a Line

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

- $\vec{r}_0$ = position vector of a known point on the line
- $\vec{v}$ = **direction vector** (parallel to the line)
- t = parameter (varies over all real numbers)



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A line in **any dimension** needs just two things: a **point** and a **direction**.

Lines in 3D

Example 3

Find vector and parametric equations for the line through $P(1, 1, 1)$ and $Q(4, 5, 6)$.

Find the direction vector PQ , then write the vector equation. Expand into parametric form.

Symmetric Equations

Symmetric Equations

If $\vec{v} = \langle a, b, c \rangle$ (all nonzero), the **symmetric equations** of the line through (x_0, y_0, z_0) are:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

If one direction component is **zero**, that coordinate is constant — write it separately. For example, if $a = 0$:

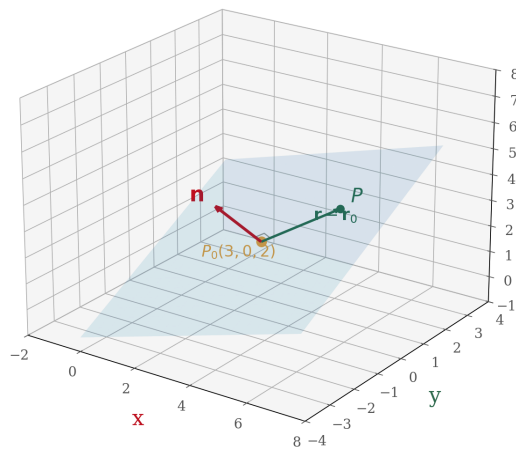
$$x = x_0 \text{ and } \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Your Turn: Line Through Two Points

Example 4

Find parametric and symmetric equations for the line through $A(2, -1, 3)$ and $B(4, 3, 1)$.

Find the direction vector AB , write parametric equations, then solve each for t .



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Equations of a Plane

- **Vector equation:** $\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$
- **Scalar equation:** $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$, where $\vec{n} = \langle a, b, c \rangle$
- **Linear equation:** $ax + by + cz = d$

The coefficients of x , y , z in the plane equation $ax + by + cz = d$ **are** the components of the normal vector:
 $\vec{n} = \langle a, b, c \rangle$.

Example: Equation of a Plane

Example 5

Find the equation of the plane through $P(3, 0, 2)$ with normal vector $\vec{n} = \langle -1, -2, 3 \rangle$.

Use the scalar equation with the given point and normal vector.

Line-Plane Intersection

Example 6

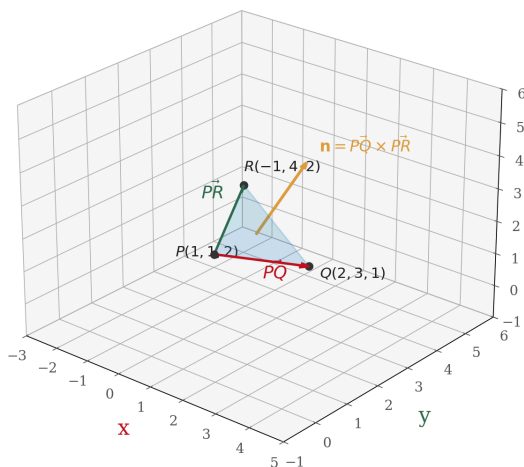
Where does the line $\vec{r} = \langle 1, 2, -1 \rangle + t\langle 4, -1, 3 \rangle$ intersect the plane $-x - 2y + 3z = 3$?

Substitute the parametric equations into the plane equation and solve for t .

Plane Through Three Points

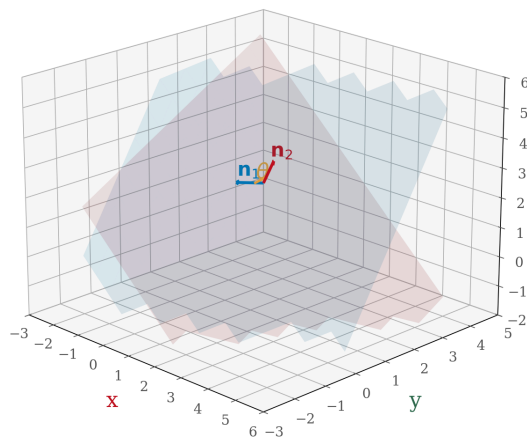
Example 7

Find the equation of the plane through $P(1, 1, 2)$, $Q(2, 3, 1)$, and $R(-1, 4, 2)$.



Find $\vec{PQ}$ and $\vec{PR}$, compute their cross product, then use the result as the normal vector.

Angle Between Planes



Angle Between Planes

If two planes have normal vectors $\vec{n}_1$ and $\vec{n}_2$, the angle θ between the planes satisfies:

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|}$$

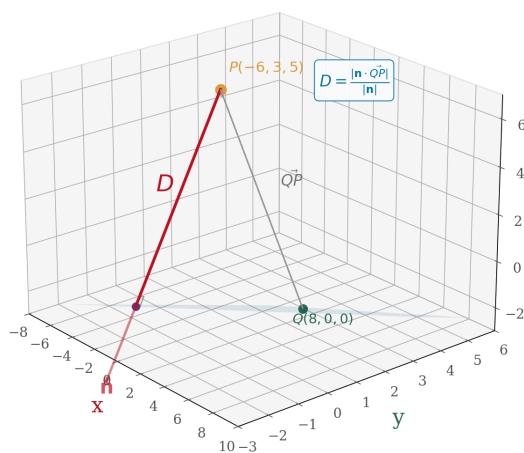
We use the **absolute value** of the dot product so that $0 \leq \theta \leq 90^\circ$.

Example 8

Find the angle between the planes $x - 2y + z = 2$ and $3x - 2y + 3z = 3$.

Read off the normal vectors, then use the angle formula.

Distance from a Point to a Plane



Distance from a Point to a Plane

The distance from point $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz = d$ is:

$$D = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Example 9

Find the distance from $P(-6, 3, 5)$ to the plane $x - 2y - 4z = 8$.

Method: pick any point Q on the plane, form PQ , then project onto the normal. Or use the formula.

Geometry True/False — Lines and Planes in 3D

Think carefully about each one. Sketch examples and counterexamples.

Homework

Section 12.5: 2, 9, 10, 16, 21, 25, 30, 33, 57, 64, 71